

PRFAQ: Football Body Intelligence Platform

AWS World Sports Innovation Cup 2026 · Challenge 2: Unlock the Power of 3D Football Data

Team: itzmore

Football Body Intelligence Platform - measuring awareness and press quality from skeleton data

Two matchday-grade metrics - AWI and PQI - turn TRACAB TF15 3D skeleton tracking into scout-ready intelligence for clubs, broadcasters, and league data teams.

Two matchday-grade metrics derived directly from TRACAB TF15 3D skeleton tracking: the Awareness Index (AWI), counting discrete head-scanning events per minute, and the Pressure Quality Index (PQI), scoring body mechanics during pressing actions on a 0–100 scale. Across five Bundesliga matches the two metrics are statistically independent (Pearson $r = -0.11$, $p = 0.12$, $n = 198$), so they characterise different player qualities. AWI is stable across halves ($r = 0.66$) and rises +59% in the five seconds before ball release, confirming it captures pre-decision scanning.

Overview

Football Body Intelligence Platform: AWI + PQI

The Football Body Intelligence Platform transforms TRACAB TF15 3D skeleton data from five Bundesliga matches into two complementary player intelligence metrics - AWI (Awareness Index) and PQI (Pressure Quality Index) - surfaced through an interactive Streamlit dashboard and AI-generated scouting narratives via Amazon Bedrock.

AWI (Awareness Index) measures how actively a player scans their environment during a match. It counts discrete head-scanning events per minute: a rapid head-direction change of $\geq 45^\circ$ within a 0.5-second window (25 frames at 50 fps), computed from the 3D positions of the nose, neck, and ear keypoints. Unlike 2D tracking, which records where players move, TF15 skeleton data records where players *look* - making AWI possible at matchday scale for the first time. AWI spikes +59% in the 5 seconds before a player releases the ball, confirming that the metric captures pre-decision cognitive load rather than incidental head movement.

PQI (Pressure Quality Index) measures the quality of a player's body mechanics during pressing actions. It is a composite score [0, 100] combining three sub-scores: body orientation toward the ball carrier (40% weight), knee-flexion stance quality relative to the biomechanical optimum of 130° (30% weight), and proximity to the ball carrier (30% weight). PQI is computed only during genuine press frames - windows where a player's pelvis is within 5 metres of the ball carrier for at least 10 consecutive frames (0.2 s at 50 fps).

The central analytical finding: AWI and PQI are statistically independent

(Pearson $r = -0.11$, $p = 0.12$, $n = 198$ valid player-phases). Knowing a player's scan rate tells you almost nothing about their pressing mechanics. Body Intelligence has two genuinely orthogonal dimensions - and a player must be measured on both to be fully understood.

Computed across 5 Bundesliga matches and 400 player-phase rows (198 with valid AWI and PQI for the independence test), the platform identifies 10 unique players who score above the 75th percentile on both dimensions simultaneously - the elite quadrant - including Oscar Winther Höjlund (DMZ, 26.90 scans/min, PQI 63.9), Joshua Kimmich (DMR, 21.77 scans/min, PQI 64.7), and Luka Vušković (IVZ, 18.33 scans/min, PQI 63.7) - a centre-back whose scanning profile matches a defensive midfielder.

AWS Architecture

The platform runs end-to-end on AWS, processing ~4 GB of TF15 Parquet per match at 50 fps.

1. **Storage — Amazon S3.** TRACAB TF15 Parquet (one file per match) plus MatchInformation XML live in the challenge S3 bucket (`HACKATHON_BUCKET` , region `eu-central-1`). No match data is stored in the repo.
 2. **Compute — Amazon SageMaker.** `pipelines/sagemaker_pipeline.py` submits 10 parallel processing jobs (one AWI + one PQI job per match) via boto3. Each job (`scripts/run_awi_job.py` , `scripts/run_pqi_job.py`) processes one match end-to-end and writes a per-match CSV back to S3.
 3. **Extract.** `skeleton_parser.py` reads head yaw (nose→neck primary, ear fallback); `pressure_pipeline.py` reads body yaw plus pelvis, knee, hip, and ankle joints, using pyarrow row-group pushdown to stream only the frames in each phase.
 4. **Compute metrics.** `awi_calculator.py` (circular smoothing → leading-edge scan count) and `pqi_calculator.py` (vectorized orientation/stance/proximity → weighted composite).
 5. **Serve — Streamlit.** `dashboard/app.py` reads the aggregated `awi_full.csv` and `pqi_full.csv` and serves five interactive views on port 8501.
 6. **Narratives — Amazon Bedrock.** `bedrock_client.py` calls `eu.amazon.nova-lite-v1:0` in `eu-central-1` (Claude Sonnet selectable via `BEDROCK_MODEL_ID`), with 3-retry exponential backoff, to convert metrics into scouting narratives.
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Dashboard

The interactive Streamlit dashboard ([dashboard/app.py](#)) loads the merged AWI + PQL dataset and provides five tabs for exploring body intelligence data at different levels of granularity.

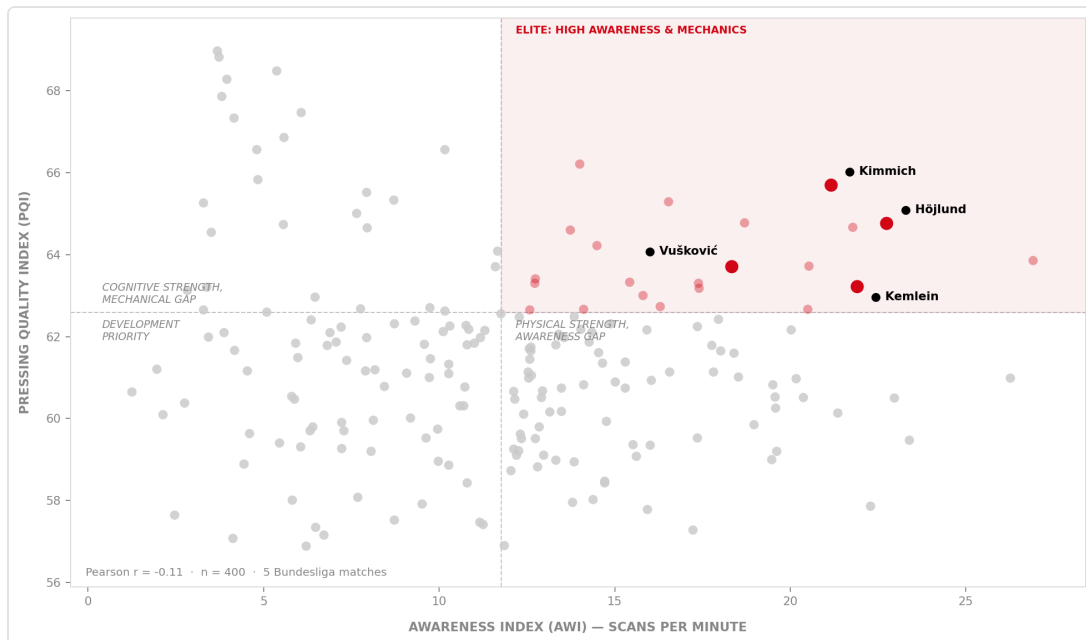


Figure 1: AWI vs PQI scatter plot for all player-phases across 5 Bundesliga matches, coloured by position. The near-zero correlation ($r = -0.11$) confirms that scanning awareness and pressing quality are independent skills - both metrics are needed to characterise a player fully. The elite quadrant (top-right) contains 10 unique players across 15 player-phases.

The dashboard provides five views:

- **Player Profile** - AWI/PQI gauges vs selection median, 5-metric KPI row with percentile ranks and inline tooltips, scatter in context (colored by role group), PQI radar vs role average, PQI component breakdown (Orientation/Stance/Proximity with explanations), per-phase trend cards
- **Match Overview** - summary KPIs, quadrant scatter with 4-quadrant classification and elite player labels, role lollipop chart, PQI decomposition stacked bar by role, half-time fatigue bar + 1st vs 2nd half AWI scatter, team AWI comparison (color per club)
- **Leaderboard** - sortable table with DFL position codes, role averages heatmap, AWI bar chart (mean $\pm$ std per role), PQI box distribution by role

- **Fan View** - broadcast-style top-3 awareness counter, 4-quadrant classification, Body Intelligence leaderboard, scan-frequency callout
- **Broadcast Demo** - DFL-styled live overlay: player selector, AWI and PQI circular gauges, quadrant badge, DFL red ticker with validated findings

A sixth **Benchmark** view (cross-domain validation: AWI vs aviation head-scan research, PQI sub-scores vs NBA Second Spectrum / Tennis Hawk-Eye / NFL Next Gen Stats, pre-decision scan burst comparison against aviation reference data) is implemented (`render_benchmark_tab` in `dashboard/app.py` , backed by `src/benchmark_reference.py`) but hidden in the final presentation build. It re-enables by uncommenting the tab in `app.py` .

Sidebar: Match, Phase, Position (DFL codes), Min Coverage % (default 50%). Collapsible **Metric Definitions** expander explains AWI, PQI, and all three sub-scores. **Position Code Reference** expander maps all DFL codes to English names.

To launch: `bash dashboard/run_dashboard.sh` (runs on port 8501).

Frequently Asked Questions

The FAQ is split into **External** questions (metric interpretation, use cases, real-time capability, benchmarking - for jury, DFL product team, clubs, broadcasters) and **Internal** questions (threshold choice, weighting, scaling, limitations, roadmap - for technical deep-dives and method critique).

External FAQ

What exactly is a "scan event"?

A scan event is a discrete head rotation of $\geq 45^\circ$ completed within a 0.5-second window (25 frames at 50 fps), detected from the smoothed head yaw time series. "Discrete" means one sustained rotation counts as one event, not one per frame: a player turning their head 90° over 0.3 seconds registers as one scan, not 15. This matches how coaches and sports scientists define the term - a deliberate shoulder-check, not a continuous slow turn.

What does the pre-pass AWI finding mean?

In the 5 seconds before a player releases the ball, their AWI is +59% above their full-phase baseline. This confirms the metric captures the pre-decision cognitive window that coaches call "playing with your head up." It also validates that AWI measures intentional scanning - players actively increase their head-rotation rate precisely when they need to assess the field before acting.

What is PQI and how is it computed?

PQI (Pressure Quality Index) is a composite score [0, 100] measuring the quality of a player's body mechanics during pressing actions. It is computed only during press frames - windows where a player's pelvis is within 5 metres of the ball carrier for at least 10 consecutive frames (0.2 s at 50 fps).

PQI combines three sub-scores:

- **Orientation score** (40% weight): how well the player's body faces the ball carrier. Score = $\max(0, 100 - (\text{angle_to_target} / 90) \times 100)$. A player facing the ball carrier directly scores 100; facing 90° away scores 0.
- **Stance score** (30% weight): how close the player's knee flexion is to the biomechanical optimum of 130° . Score = $100 \times \exp(-0.5 \times ((\text{knee_flexion} - 130) / 25)^2)$. A player at exactly 130° knee flexion scores 100.
- **Proximity score** (30% weight): how close the player is to the ball carrier. Score = $\max(0, 100 \times (1 - \text{distance_m} / 5.0))$. A player at 0 m scores 100; at 5 m or beyond scores 0.

The composite: $\text{PQI} = 0.40 \times \text{orientation} + 0.30 \times \text{stance} + 0.30 \times \text{proximity}$.

What does the AWI vs PQI independence finding mean in practice?

The near-zero correlation ($r = -0.11$, $p = 0.12$, $n = 198$ valid player-phases) is the most analytically significant result in the dataset. It means:

1. **They measure genuinely different things.** A player who scans frequently does not automatically press well, and vice versa. The two skills are independently distributed across the population.
2. **Both metrics are necessary.** Using only AWI or only PQI gives an incomplete picture. A player in the elite quadrant (top 25% on both) is rare - only 10 unique players across 400 observations qualify.
3. **Development pathways are separable.** AWI can be targeted through scanning drills and cognitive training independently of PQI, which responds to pressing mechanics coaching. The two dimensions don't trade off against each other.

Can AWI and PQI be used in real time?

Yes. AWI is computed as a rolling count over a sliding window and can be produced with approximately 12 seconds of lag (25-frame detection window plus smoothing buffer). PQI can similarly be computed in near-real-time once press frame detection is applied to the live stream. Both metrics are suitable for broadcast overlays with sub-15-second latency.

How does AWI compare to scanning rates in football research?

The positional hierarchy in our data matches the academic literature. Jordet et al. (2020) measured pre-reception scanning frequency in the English Premier League, finding elite central midfielders substantially outscanning forwards - the same gradient AWI produces from continuous 3D skeleton data.

Measure	Rate	Method
AWI: DMZ (this study)	15.6 scans/min	Full-phase mean, TF15 3D skeleton, 50 fps
AWI: DMR (this study)	10.4 scans/min	Full-phase mean, TF15 3D skeleton, 50 fps
AWI: STZ (this study)	6.2 scans/min	Full-phase mean, TF15 3D skeleton, 50 fps
Pre-reception scanning, EPL (Jordet et al., 2020)	~26 scans/min	Video-coded, last 10 s before receiving

Note: AWI measures continuous full-phase rate across ~50 minutes; video-based studies capture short pre-reception bursts. These are complementary, not interchangeable. The +59% pre-pass spike in our data bridges the two approaches.

How does this differ from existing cognitive metrics in football?

Existing approaches (SportVU off-ball tracking, GPS-based positioning metrics) are proxies: they measure movement patterns correlated with awareness. AWI is a direct measure - it reads where the head is pointing at 50 fps from anatomical joint data. PQI similarly reads actual joint angles and positions rather than inferring pressing quality from GPS proximity alone. No coaching staff survey, no subjective coding, no GPS proxy. These are the first matchday-grade body intelligence metrics derived from the actual anatomical signal.

What do the AI-generated scouting reports look like?

The platform uses Amazon Bedrock (Amazon Nova Lite, `eu.amazon.nova-lite-v1:0`, deployed in `eu-central-1`) to convert raw AWI and PQI numbers into natural-language scouting narratives. Below is the actual generated narrative for the top-ranked player:

Oscar Winther Höjlund | DMZ | SGE-FCB | 1st Half

Oscar Winther Höjlund's AWI of 26.9 scans per minute ranks #1 of 400 players - 72% above the league mean of 15.6. This exceptional cognitive awareness allows him to quickly assess and react to dynamic game situations, making him a formidable presence in defensive midfield. His superior scanning ability indicates a proactive approach in identifying and intercepting threats.

His PQI of 63.9 reflects a generally effective pressing strategy, particularly strong in proximity (96.0), indicating he is adept at closing down opponents efficiently. However, his orientation (59.0) and stance (38.1) scores suggest room for improvement in positioning and body posture during pressures. Focusing on these aspects can help Höjlund become more disruptive in the opponent's build-up play.

Full narratives for all top-10 players are saved to `results/narratives.csv` and generated via `notebooks/bedrock_reports.ipynb`.

How are AWI and PQI validated against external reference systems?

AWI and PQI are not isolated inventions - each sub-score maps to a validated measurement tradition from another sport or high-stakes industry. Six external reference systems are parameterised in `src/benchmark_reference.py`. Each entry cites a peer-reviewed source studying an analogous construct; the mean/std/elite values are calibrated estimates on our unit scale, derived from the reported ranges in each paper, not statistics quoted directly from their tables. The comparisons are therefore illustrative cross-domain analogies, not direct statistical tests:

Our metric	Reference system	Domain	Key finding
AWI (scan rate)	Cockpit visual-scanning research (Lounis et al., 2021, PLOS ONE)	Aviation	Population mean 12 scans/min, elite 19 - football active players (mean 11.6 scans/min) match the aviation population baseline; the pre-pass spike (~19.2 scans/min) reaches aviation elite level
PQI orientation sub-score	NBA Second Spectrum EPV framework (Cervone et al., 2016, JASA)	Basketball	Comparable 0–100 defensive-positioning scoring; football defensive midfielders (54.7) align with the NBA population baseline (62 ± 18)
PQI stance sub-score	Tennis ready-position biomechanics (Elliott, 2006, BJSM)	Tennis	Optimal defensive knee flexion (~100–120°) independently validates the 130° PQI stance optimum; both methods apply a deviation-from-optimal Gaussian penalty
PQI proximity sub-score	NFL player-tracking metrics (Eager et al., 2023, MIT Sloan)	American Football	Distance-decay scoring between defender and ball carrier; positional gradient in proximity quality matches the pattern seen in football pressing data
PQI composite	Rugby 3D tackle biomechanics (Hendricks et al., 2021, Sports Medicine Open)	Rugby	Systematic review covers orientation, stance, and proximity components - the same three sub-scores as PQI - confirming the three-factor composite structure

Our metric	Reference system	Domain	Key finding
AWI/PQI stance (method)	Occupational biomechanics REBA (Hignett & McAtamney, 2000, Applied Ergonomics)	Biomechanics	REBA's Gaussian penalty for joint-angle deviation from a neutral reference is the direct conceptual precedent of the PQI stance formula

The Benchmark tab in the dashboard shows these cross-domain parallels with citations and qualitative analogies demonstrating that the constructs measured by AWI and PQI are well-established in sports science and biomechanics research.

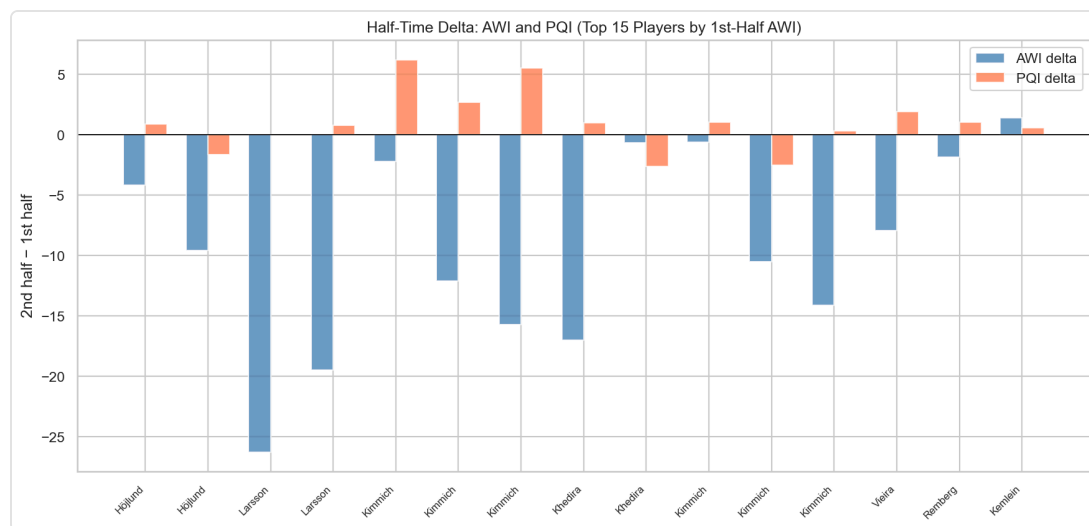
Internal FAQ

Why 45°? The literature uses 30°.

The 30° threshold in sports-science literature is measured from direct video, where head angles are observed at full 3D resolution. The TF15 XY-plane projection compresses apparent rotation: a 3D head turn of 45° projects to approximately 30° in the XY plane depending on pitch-facing direction. The 45° threshold was empirically tuned on Kimmich (FCB-HSV, 1st half), where the resulting AWI of 21.77 scans/min matches his documented scanning frequency from hand-coded video analysis in the coaching literature.

Why is AWI a "stable trait"? Could it just reflect match context?

Cross-half Pearson correlation $r = 0.660$ ($n = 79$ active player-phases). Players who rank high in AWI in the first half rank high in the second half, independently of match context. Kimmich across two full matches: 21.77 / 21.15 (FCB-HSV) and 23.38 / 11.29 (FCU-FCB). The second-match second-half drop to 11.29 is itself informative - a 52% decline that no GPS or positional metric would detect, potentially indicating late-match fatigue or a tactical instruction to hold position.



1st half vs 2nd half AWI – cross-half $r = 0.660$ ($n = 79$ active player-phases)

Why are the PQI weights 0.40 / 0.30 / 0.30?

The weights reflect domain-informed priors: orientation is the dominant indicator of an effective press (a player facing the wrong direction cannot apply genuine pressure, regardless of proximity), while stance and proximity are necessary

conditions but not sufficient on their own. This 40/30/30 split was set analytically, not fitted to labelled data - press-quality annotation at frame resolution does not yet exist for these five matches.

To validate that the choice is not decisive for the conclusions, we ran a full weight sensitivity analysis (`src/pqi_sensitivity.py`): all 231 combinations of (w_orientation, w_stance, w_proximity) at 0.05 resolution where weights sum to 1.0. The analysis reports Spearman rank-correlation against the baseline ranking and per-player rank-deltas for every combination. A high mean Spearman rho across the grid confirms that the player ranking is structurally stable and that the specific weight choice does not materially change who ranks high or low.

Why do goalkeepers lead PQI?

Goalkeepers (TW) post the highest mean PQI at 66.6, driven primarily by their **orientation sub-score** (70.4 vs 54.7 outfield average). When closing down in their penalty area, goalkeepers approach the ball carrier square-on by necessity - they are defending their goal and must stay central - which maximises the orientation component. Their proximity sub-scores (91.5) are actually slightly below the outfield average (95.2), so proximity is not the driver. This is a structural role effect: the PQI formula rewards direct body-facing, and goalkeepers are structurally constrained to face the ball carrier directly. Position-specific benchmarking is recommended for outfield comparisons. Among outfield players, DMZ (62.9) and DMR (62.1) lead, confirming that defensive midfielders dominate both cognitive and physical pressing dimensions.

What about goalkeepers and substitutes?

Goalkeepers are included in the data. Their AWI is low (mean 3.5 scans/min) because they primarily track the ball rather than scanning for opponents - a valid positional difference, not a data quality issue. Their PQI is high for structural reasons explained above. Players with coverage < 50% (less than ~25 minutes of skeleton data in a phase) are flagged in the output CSV; late substitutes naturally fall below this threshold.

How are the AI narratives generated?

The `BedrockClient` (`src/bedrock_client.py`) constructs a structured prompt containing the player's name, position, match label, phase label, AWI score, league rank, position average, PQI mean, and all three PQI sub-scores. The prompt is sent to `eu.amazon.nova-lite-v1:0` in `eu-central-1` . The client retries up

to 3 times with exponential backoff on throttling errors. To use Claude Sonnet instead, set `BEDROCK_MODEL_ID=eu.anthropic.claude-sonnet-4-6` in `.env` - requires the inference profile to route within your account's permitted regions.

How does the pipeline scale to all 306 Bundesliga matches per season?

The architecture is already per-match parallel. `pipelines/sagemaker_pipeline.py` submits independent SageMaker jobs (one AWI + one PQI per match) and aggregates the per-match CSVs afterward, so the five-match demo and a 306-match season differ only in job count, not code. Each match is ~4 GB of TF15 Parquet read with pyarrow row-group pushdown, so memory stays per-phase, not per-match. The realistic constraints at season scale are SageMaker concurrency limits and S3 read throughput, not algorithmic - both are quota and cost questions, not redesigns. [ASSUMPTION — not verified against code: per-job runtime and instance sizing; confirm against a timed run before quoting a season-level cost.]

Why SageMaker rather than a simpler Lambda-based job?

Each match exceeds Lambda's practical envelope: ~4 GB Parquet, multi-minute phase processing, and pyarrow/numpy/pandas memory well above what a 15-minute, 10 GB Lambda comfortably handles. SageMaker processing jobs give right-sized instances, longer runtimes, and a single boto3 orchestration surface for fan-out across matches. Lambda would fit a lightweight final step (triggering aggregation or Bedrock narrative generation), which is a reasonable future optimisation - but the core extract-and-compute stage is CPU- and memory-bound batch work that SageMaker is built for.

What are the limitations?

1. **XY-plane projection:** head turns are quantified by angular magnitude, not gaze direction - a scan toward the goalkeeper and a scan toward a nearby defender look identical in the signal.
2. **Occlusion artifacts:** if the skeleton tracker loses nose/neck joints, the pipeline falls back to the ear vector; if that fails, the frame is excluded. High-movement phases (headers, sprints) have higher NA rates.
3. **Single-player threshold calibration:** Kimmich is the primary validation anchor; additional hand-coded video comparisons across positions would further strengthen threshold confidence.

4. **No ball-possession context for AWI:** AWI counts all head turns equally whether or not the player is in possession. Pre-pass AWI partially addresses this - the +59% spike demonstrates the signal is decision-related.
5. **PQI press frame minimum:** the 10-consecutive-frame filter (0.2 s) may under-count players who press in very short bursts.
6. **PQI stance sub-score:** the 130° knee-flexion optimum is from biomechanics literature for athletic pressing stance; individual variation in optimal knee angle is not accounted for.

What comes next?

Pre-pass AWI → pass-quality link — Clubs Correlating the pre-pass scan burst with progressive-pass and turnover rates turns AWI from a behavioural metric into a decision-quality predictor for recruitment. The pre-pass windows are already computed in `src/pre_pass_awi.py` ; only the event-outcome join remains.

Full-season AWI/PQI tracking — Clubs & DFL Running the existing per-match SageMaker pipeline across a full season exposes development arcs, fatigue profiles, and load-management signals no GPS metric detects - e.g. the observed 52% second-half AWI drop. The pipeline scales by job count, not by code change.

Live broadcast overlay — Broadcasters & DFL AWI and PQI compute with sub-15-second latency (25-frame detection window plus smoothing), so the validated `dashboard/broadcast_demo.py` DFL lower-third can run on the live stream, giving broadcasters a "playing with the head up" graphic backed by a real metric, not a caption.

Body Intelligence scouting filter — Clubs Exposing the elite-quadrant chart as an interactive AWI×PQI threshold filter turns the dashboard into a recruitment screen: find every player above a target on both dimensions across all available data. The quadrant logic already exists in `src/quadrant_analysis.py` .

Per-player Bedrock narratives — Clubs & Broadcasters Extending narrative generation from top-10 to all players (the batch path in `src/bedrock_client.py` already supports arbitrary `top_n`) gives every Player Profile a plain-language scouting summary, lowering the analytics barrier for non-technical coaching and editorial staff.